Flow Shop Scheduling

Sensitivity: Internal

Scheduling - Single Stage

Consider a set of jobs $\{1,2, \dots, n\}$ which need to be processed by a single machine/stage/activity.

e.g. a supermarket queue at one checkout.

Makespan – Total time taken to process all jobs

Completion Time – the time at which a job is finished: its waiting time in the queue plus its own processing time.

e.g. You are buying a Kit-Kat and are behind someone with a full trolley.

If they are kind enough to let you through, what happens to the **Makespan** and the **Completion Time** ?

Makespan isn't concerned with order. The checkout takes the same amount regardless of order.

Completion Time however is vastly different. If you are not let by, you both wait for almost the entire **Makespan**.

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Shortest Processing Time

Consider a set of jobs $\{1,2,\dots,n\}$ which need to be processed by a single machine (stage).
e.g. a supermarket queue at one checkout.

C_{max} - Makespan, i.e. the last completion time
C_j - Completion time of job j
$\sum_j C_j$ - Sum of all completion times (total completion time).
p_j - length of job j

Shortest processing time, order the jobs in ascending order

$$p_1 \leq p_2 \leq \cdots \leq p_n$$



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Shortest Processing Time

Lemma 1: Let $\boldsymbol{a}$ be a vector of descending order and $\boldsymbol{b}$ a vector of the same length of no specific order. The minimum scalar product of $\boldsymbol{a} \cdot \boldsymbol{b}$ occurs when $\boldsymbol{b}$ is of ascending order.

Proof:

$$\boldsymbol{a} = (a_1, \dots, a_n), \qquad \boldsymbol{b} = (b_1, \dots, b_n)$$

Such that:

$$a_1 \geq a_2 \geq \cdots \geq a_n$$

Assume we have two indices $i < j$ (not necessarily adjacent) such that:

$$b_i > b_j$$

Define a new vector $\boldsymbol{b'}$, where these two elements are swapped.

$$\boldsymbol{a} = \left(\dots, a_i, \dots, a_j, \dots\right), \qquad \boldsymbol{b} = \left(\dots, b_i, \dots, b_j, \dots\right), \qquad \boldsymbol{b'} = \left(\dots, b_j, \dots, b_i, \dots\right)$$

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Shortest Processing Time

$$\boldsymbol{a} \cdot \boldsymbol{b'} = \cdots + \cdots + a_i b_j + \cdots + a_j b_i + \cdots$$

$$\boldsymbol{a} \cdot \boldsymbol{b} = \cdots + \cdots + a_i b_i + \cdots + a_j b_j + \cdots$$

$$\begin{aligned} \boldsymbol{a} \cdot \boldsymbol{b'} - \boldsymbol{a} \cdot \boldsymbol{b} &= a_i\left(b_j - b_i\right) + a_j\left(b_i - b_j\right) \\ &= \left(a_i - a_j\right)\left(b_j - b_i\right) \end{aligned}$$

But $a_i \geq a_j \Rightarrow a_i - a_j \geq 0$ and $b_i > b_j \Rightarrow b_j < b_i \Rightarrow b_j - b_i < 0$

$$\therefore \left(a_i - a_j\right)\left(b_j - b_i\right) \leq 0$$

$\Rightarrow$

$$\boldsymbol{a} \cdot \boldsymbol{b'} - \boldsymbol{a} \cdot \boldsymbol{b} \leq 0$$

$\Rightarrow$

$$\boldsymbol{a} \cdot \boldsymbol{b'} \leq \boldsymbol{a} \cdot \boldsymbol{b}$$

Thus, swapping any two elements in $\boldsymbol{b}$ where $b_i > b_j$ results in an equal or lower scalar product

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Shortest Processing Time

Therefore, for any $\boldsymbol{b}$ not in ascending order, there exists a swap which results in an equal or lower scalar product $\boldsymbol{a} \cdot \boldsymbol{b}$.

Thus, the minimum is attained when $\boldsymbol{b}$ is in ascending order. ■

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Shortest Processing Time

Theorem: SPT is optimal for minimising the sum of completion times $\sum_j C_j$

Proof:

$$C_1 = p_1, \qquad C_2 = p_1 + p_2, \qquad C_j = \sum_{i=1}^{j} p_i$$

$$\begin{aligned}\sum_{j=1}^{n} C_j = {} & p_1 \\ & + p_1 + p_2 \\ & + p_1 + p_2 + p_3 \\ & \vdots \\ & + p_1 + p_2 + \cdots + p_{n-1} + p_n \\ & = np_1 + (n-1)p_2 + \cdots + 2p_{n-1} + p_n \\ & = (n, n-1, \dots, 2,1) \cdot (p_1, p_2, \dots, p_{n-1}, p_n)\end{aligned}$$

$(n, n-1, \dots, 2,1)$ is of descending order, therefore, to obtain a minimum by the previous lemma $(p_1, p_2, \dots, p_{n-1}, p_n)$ needs to be of ascending order, i.e. shortest processing time first.

■

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Flow Shop Scheduling

We will deal with a simpler set up than the general job shop scheduling problem.

Often a company will have multiple jobs (e.g. products) that go through multiple stages (machines) sequentially.

Each job may require a different amount of time at each stage.

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Example

Consider a 2 stage process in sequence.



Each job requires different times at each stage.

e.g.

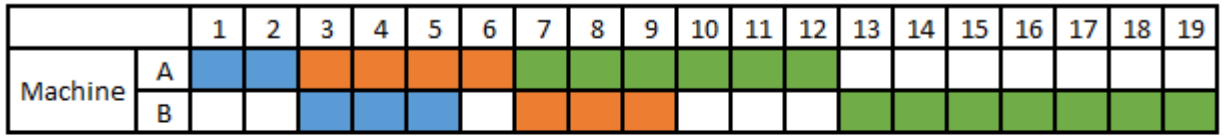
Jobs		P	Q	R
Machine	A	2	4	6
	B	3	3	7

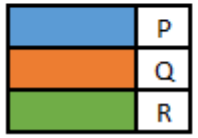
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Example

Jobs		P	Q	R
Machine	A	2	4	6
	B	3	3	7

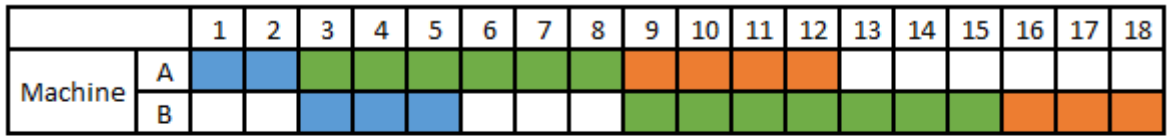
Schedule jobs in order they appear. PQR

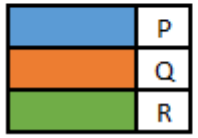




Is this optimal for Makespan? i.e. is everything completed as quickly as possible?

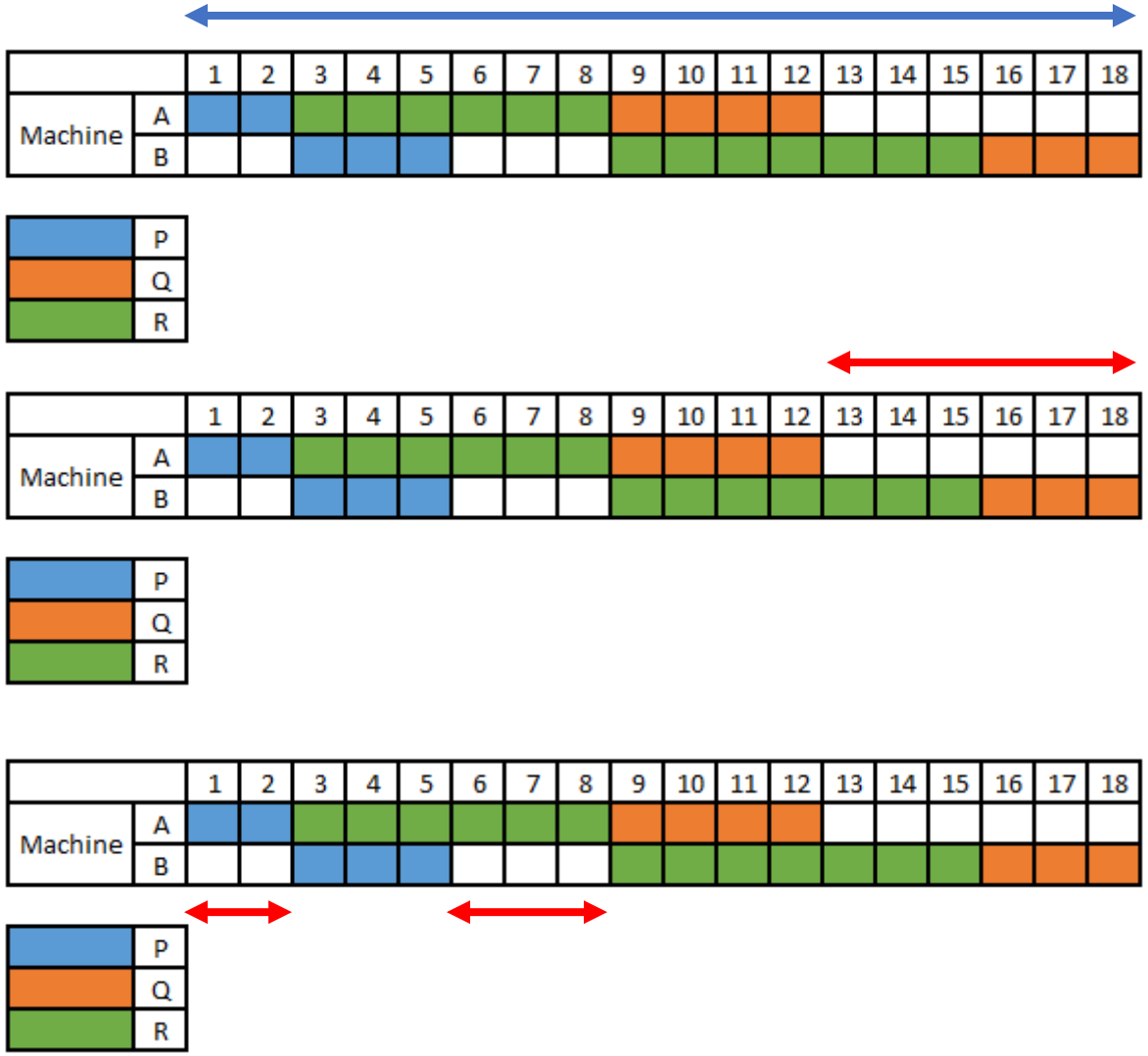
No PRQ





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Makespan and Idle Time



Makespan is the total time required to complete all jobs

Idle Time is the time a stage (machine) is idle.

Stage A is idle for 6 units of time.

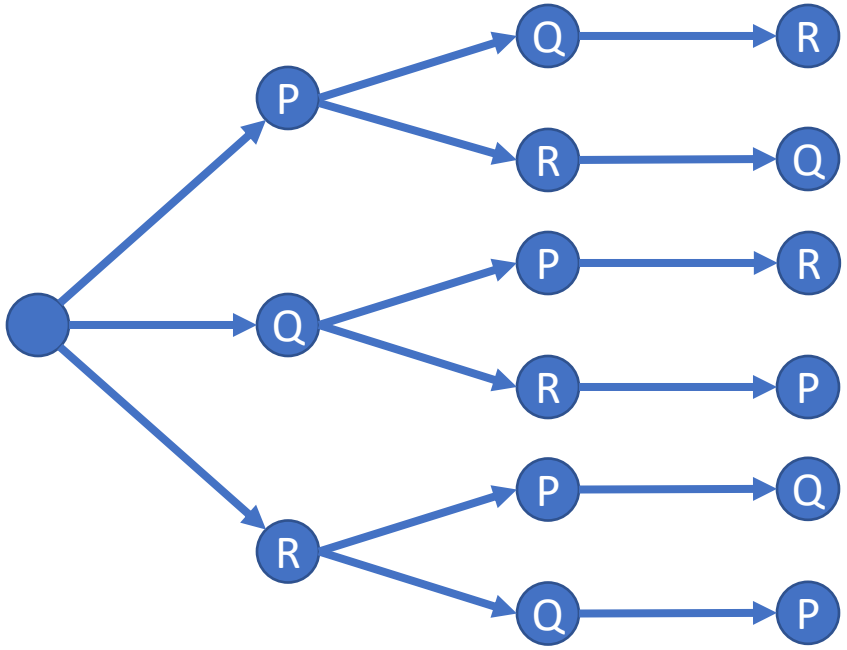
Stage B is idle for 2+3=5 units of time.

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Optimality

Optimality is difficult to check! These are **NP-Hard** problems

How many possible permutations? No repetitions and order matters!



$$3 \times 2 \times 1 = 6$$

OR

$$3! = 6$$

For n jobs there are $n!$ ways of scheduling.

e.g. 10 jobs, $10! = 3628800$

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Johnson's Rule

- Minimises Makespan
- Optimal solution for 2 stages (machines)
- Optimal solution for 3 stages (machines) given additional conditions
    - Not covering this…
- Everything else is NP-hard
- In general problems like these require clever guessing (heuristics)

It effectively schedules the jobs which are small for the first stage at the beginning, and those that are small for the second stage at the end.

https://www.rand.org/content/dam/rand/pubs/papers/2008/P402.pdf

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Johnson’s Rule (Two stage)

- Steps
  1. List the jobs
  2. Select the job with the shortest activity time, break ties arbitrarily
     - If shortest is for first stage, schedule as early as possible
     - Else if shortest is for second stage, schedule as late as possible
  3. Remove the job from list.
  4. Repeat 2 and 3 until no jobs are left to schedule

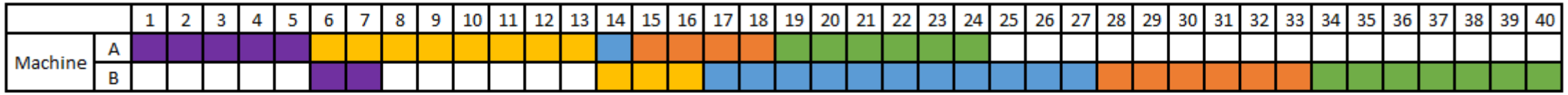
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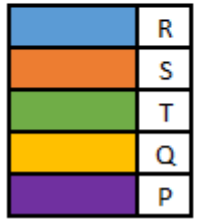
Example

Jobs		P	Q	R	S	T
Machine	A	5	8	1	4	6
	B	2	3	11	6	7

Schedule					

Before applying Johnson's Rule





Metric	Time
Makespan	40
Idle Time of A	16
Idle Time of B	11
Total Idle Time	27

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Example

1. List the jobs
2. Select the job with the shortest activity time, break ties arbitrarily
   - If shortest is for first stage, schedule as early as possible
   - Else if shortest is for second stage, schedule as late as possible
3. Remove the job from list.
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Jobs		P	Q	R	S	T
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Schedule					

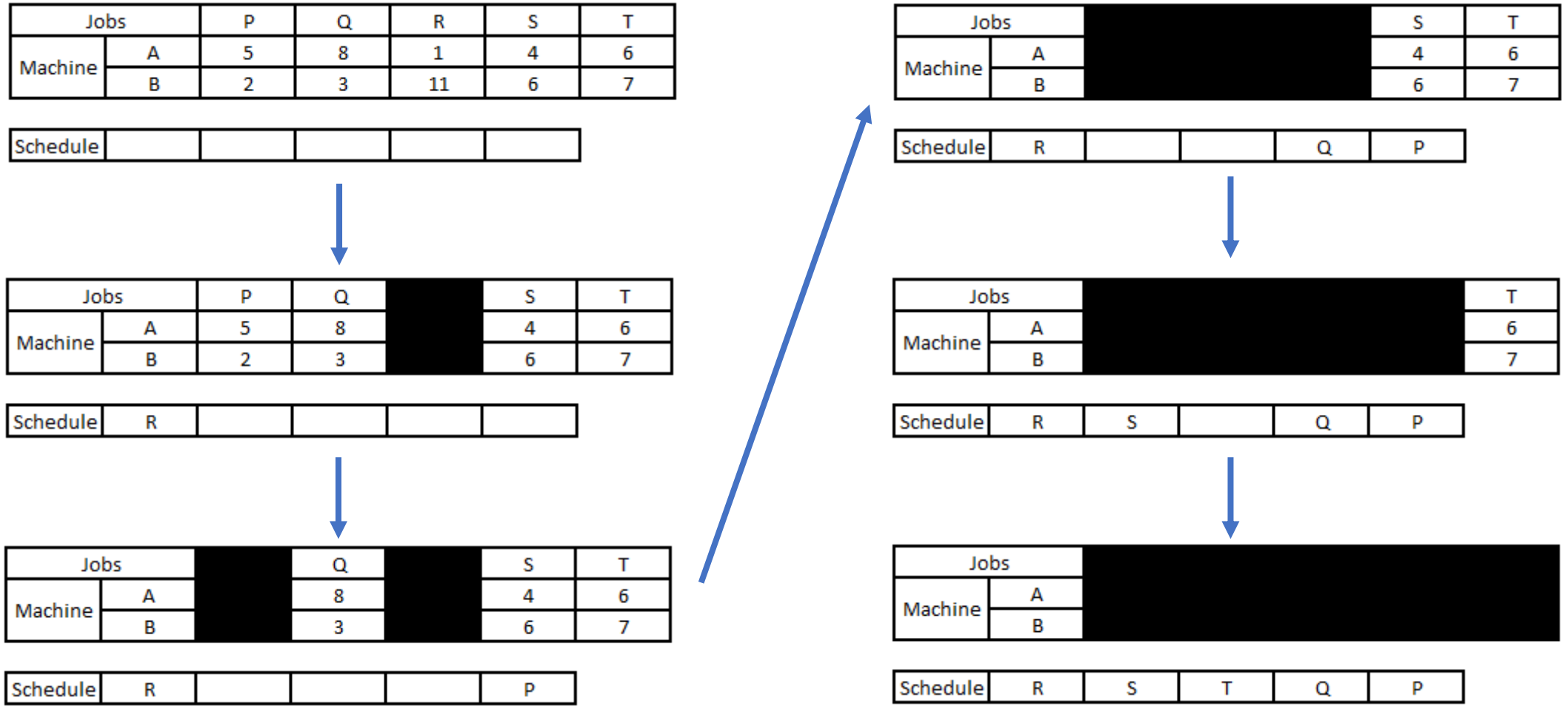


Jobs		P	Q		S	T
Machine	A	5	8		4	6
	B	2	3		6	7

Schedule	R				

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Example



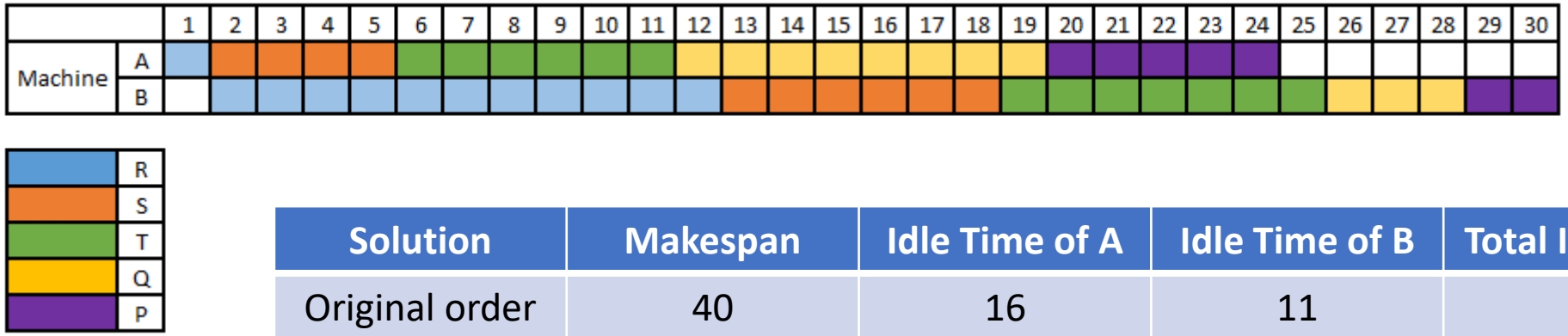
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Example

Jobs		P	Q	R	S	T
Machine	A	5	8	1	4	6
	B	2	3	11	6	7

Schedule	R	S	T	Q	P

After applying Johnson's Rule



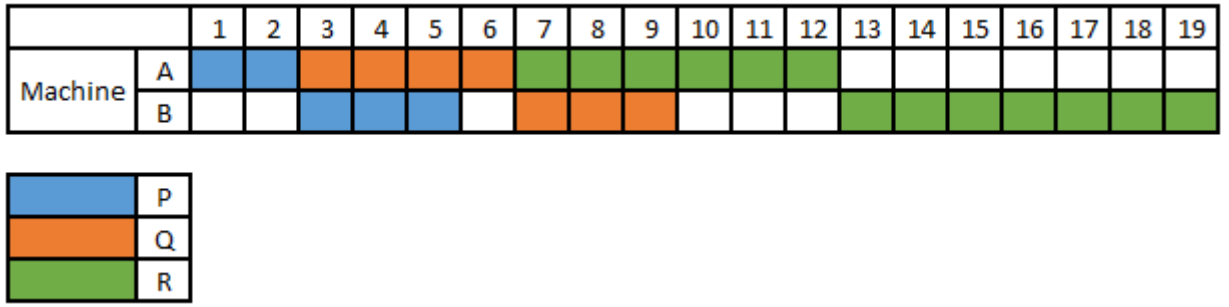
Solution	Makespan	Idle Time of A	Idle Time of B	Total Idle Time
Original order	40	16	11	27
Johnson's	30	6	1	7

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Example 2

Jobs		P	Q	R
Machine	A	2	4	6
	B	3	3	7

Schedule			



Solution	Makespan	Idle Time of A	Idle Time of B	Total Idle Time
Original order	19	7	6	13

Sensitivity: Internal

Example 2

Jobs		P	Q	R
Machine	A	2	4	6
	B	3	3	7

Schedule			

Jobs			Q	R
Machine	A		4	6
	B		3	7

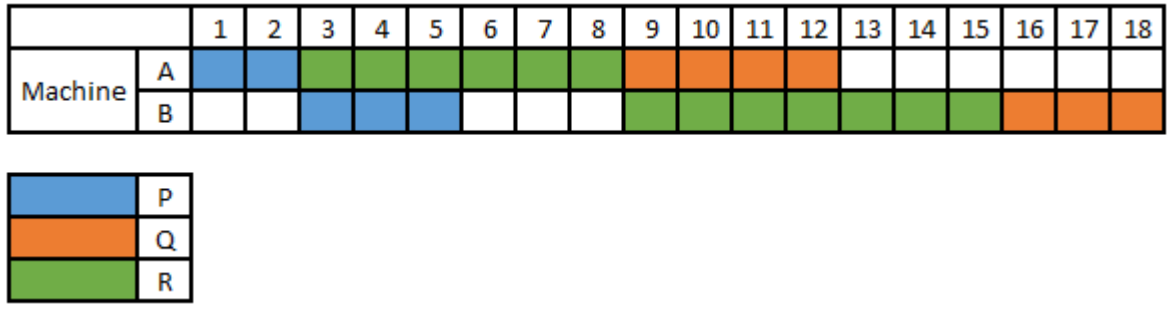
Schedule	P		

Jobs			R
Machine	A		6
	B		7

Schedule	P		Q

Jobs			
Machine	A		
	B		

Schedule	P	R	Q



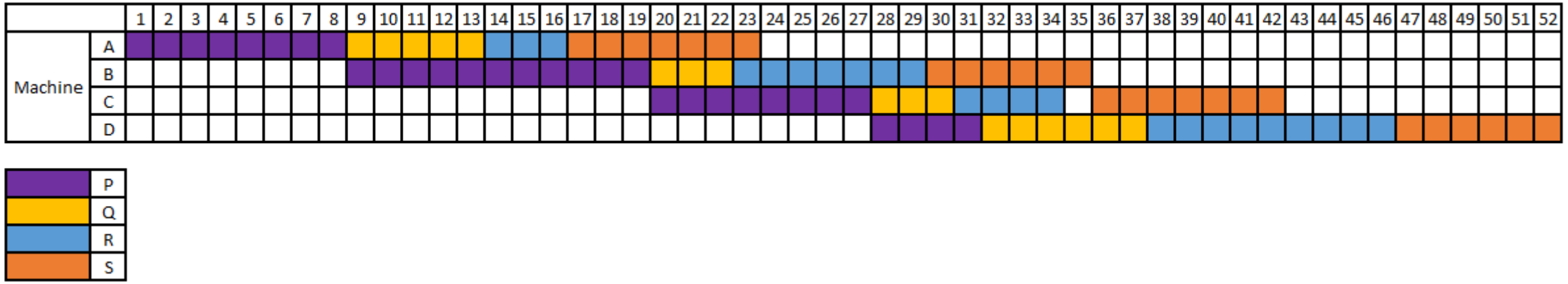
Solution	Makespan	Idle Time of A	Idle Time of B	Total Idle Time
Original order	19	7	6	13
Johnson's	18	6	5	11

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More Than 2 Machines

Jobs		P	Q	R	S
Machine	A	8	5	3	7
	B	11	3	7	6
	C	8	3	4	7
	D	4	6	9	6

Schedule				



Solution	Makespan	Idle Time of A	Idle Time of B	Idle Time of C	Idle Time of D	Total Idle Time
Original order	52	29	25	30	27	111

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More Than 2 Machines

- Johnson's rule can be used to reduce the size of the search space.
- Idea - reduce to a 2-stage problem. This is a heuristic!
  - Only produces a good solution, not necessarily optimal!
  - Often named the method/approach of Campbell, Dudek and Smith

e.g. First and last stage considered

Jobs		P	Q	R	S
Machine	A	8	5	3	7
	B	11	3	7	6
	C	8	3	4	7
	D	4	6	9	6

Schedule				



Jobs		P	Q	R	S
Machine	A	8	5	3	7
	D	4	6	9	6

Schedule	R	Q	S	P

Makespan - 46

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Other Possibilities

Aggregate the first two machines and last two machines

Jobs		P	Q	R	S
Machine	A	8	5	3	7
	B	11	3	7	6
	C	8	3	4	7
	D	4	6	9	6

Schedule				



Jobs		P	Q	R	S
Machine	A+B	19	8	10	13
	C+D	12	9	13	13

Schedule	Q	R	S	P

Makespan - 46

Aggregate the first three machines and last three machines

Jobs		P	Q	R	S
Machine	A	8	5	3	7
	B	11	3	7	6
	C	8	3	4	7
	D	4	6	9	6

Schedule				



Jobs		P	Q	R	S
Machine	A+B+C	27	11	14	20
	B+C+D	23	12	20	19

Schedule	Q	R	P	S

Makespan - 48

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For m machines

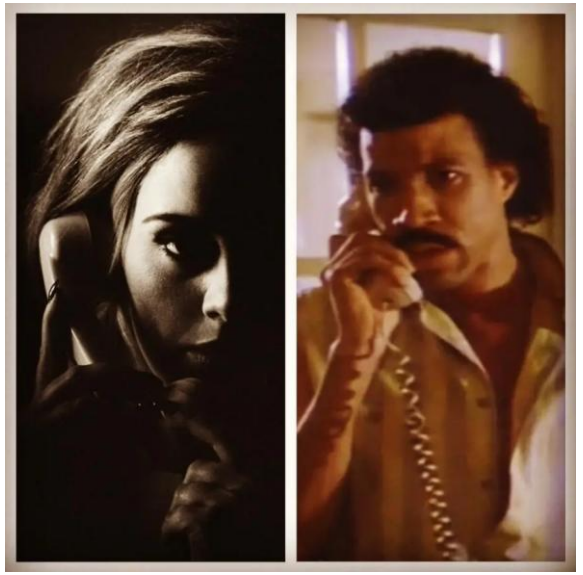
$m - 1$ possible sequences

$$
\begin{aligned}
s_1 &= 1, && m \\
s_2 &= 1 + 2, && (m-1) + m \\
s_3 &= 1 + 2 + 3, && (m-2) + (m-1) + m \\
&\vdots \\
s_{m-1} &= 1 + 2 + \cdots + (m-1), && 2 + 3 + \cdots + m
\end{aligned}
$$

Consider all these sequences and pick the best one!

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Trivia Time…



The average human body contains how many pints of blood?

- 9

The Statue of Liberty was given to the US by which country?

- France

What title is shared by a 1983 Lionel Richie song and a 2015 Adele song?

- Hello

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